

Madhav Kumar

Software Engineer | Backend & Cloud
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SUMMARY

Software Engineer at Nielsen with experience in production scale distributed systems, cloud infrastructure, and automation. Skilled in Python, Java, AWS, Kubernetes, and CI/CD, with a track record of improving reliability, performance, and operational efficiency through measurable engineering outcomes.

TECHNICAL SKILLS

Languages: Python, Java

Cloud & Infrastructure: AWS (Lambda, S3, EC2, RDS, SQS, SNS, CloudWatch, IAM), Kubernetes, Docker, Terraform

Databases: PostgreSQL, ClickHouse, MySQL

Backend & DevOps: Git, GitLab CI/CD, Microservices, REST APIs, Linux

EXPERIENCE

MTS1 | Sep 2026 – Present

Software Development Engineer Intern | Mar 2025 – Aug 2026

Audio Quality Engine

- Built a Python-based validation tool to identify noise, muffled audio, distortion, and other quality issues before recordings enter downstream processing.
- Resolved audio capture failures across 2,000+ daily pipeline runs, reducing dropouts from ~30% to <1% and restoring SLA compliance.
- Stabilized recording workflows by tuning audio buffer parameters and standardizing configurations end to end, eliminating recurring signal-loss failures.
- **Nielsen AM Tech Awards 2026 – Extra Miler:** Recognized for improving pipeline SLA, quality, and reliability through the dropout-reduction initiative.

Infrastructure & Automation

- Introduced WAV-to-FLAC compression, lowering media storage consumption by 70%, with event tracking across PostgreSQL, SNS, SQS, and ClickHouse.
- Optimized existing workflows by removing redundant execution patterns, improving pipeline processing speed by 34%.
- Automated configuration deployment across 450+ Windows VMs in multiple regions using PowerShell, ensuring consistent production environments without manual intervention.
- Engineered an AWS Lambda + CloudWatch validation workflow that reduced manual log analysis by 60% and integrated automated checks into CI/CD releases.

PROJECTS

Kubernetes Logging System | Kubernetes, Fluentd, ClickHouse

- Designed a logging setup where each running service automatically sends its logs to a central
- store with no manual collection needed.
- Tested with 50,000 plus log events, all searchable and retrievable in under a second.

GitForgeX – Version Control System | Python, Docker, REST APIs

- Developed a Git-like version control system using containerized services for repository management and version history.
- Implemented API-driven service communication, secure push handshakes, and snapshot-based storage for efficient history retrieval.

EDUCATION

Lovely Professional University – B.Tech, Computer Science & Engineering
2022 – 2026 | 87.90% | Jalandhar, Punjab, India